OneLP Capstone

Proof-of-Concept Deliverables

AI document-intelligence pipeline + portfolio analytics for Limited Partners

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Three deliverables — all targets met

DELIVERABLE 1

Extraction Accuracy

0.968

Hybrid macro F1

✓ Target ≥ 0.90 · critical-field 0.968 (≥ 0.95)

DELIVERABLE 2

Forecast Backtest

5.0%

MAE vs actual

✓ Target < 15% · 1,000-path Monte Carlo

DELIVERABLE 3

Pipeline & Latency

24s

End-to-end P50

✓ Target < 30s · P95 29s · ECE 0.056

CONTEXT

The problem & the system

The problem

LPs & family offices manage private-market portfolios across GP portals, PDFs, emails and spreadsheets — no standard schema.

- 30–40 hours/month of manual reporting
- Missed capital calls & compliance risk
- No real-time exposure / liquidity view

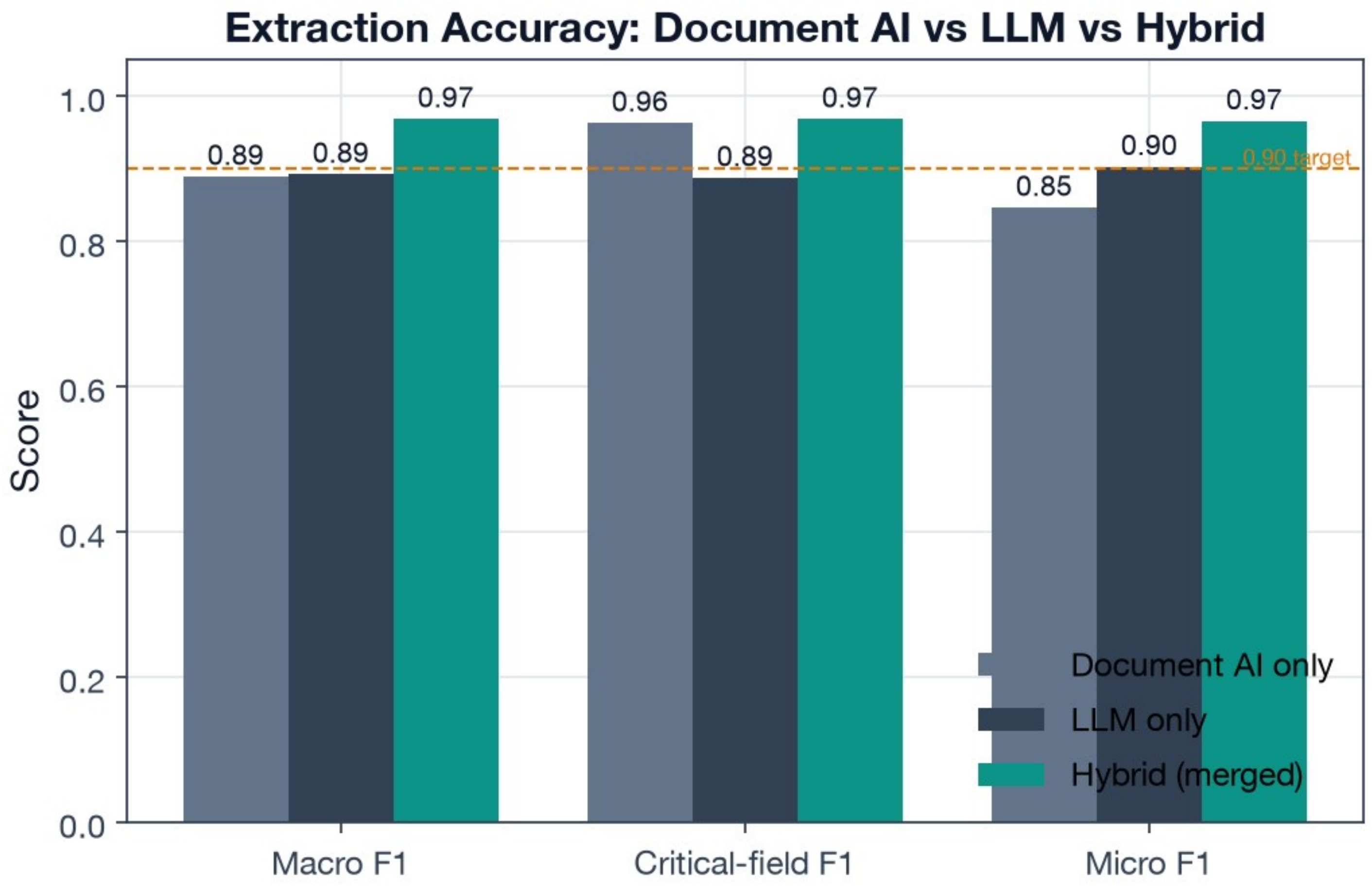
The OneLP system (3 layers)

1. Ingestion — 2-stage classify, hybrid OCR+LLM extract, merge, validate, auto-apply
2. Analytics — Monte Carlo forecasting, J-curve profiles, stress tests, liquidity reserves
3. Automation — alerts, daily briefing, scheduled reports, AI email drafting

This capstone evaluates the ingestion pipeline (D1, D3) and the analytics engine (D2).

DELIVERABLE 1 · EXTRACTION ACCURACY

Hybrid extraction clears both accuracy targets



0.968 Hybrid macro F1

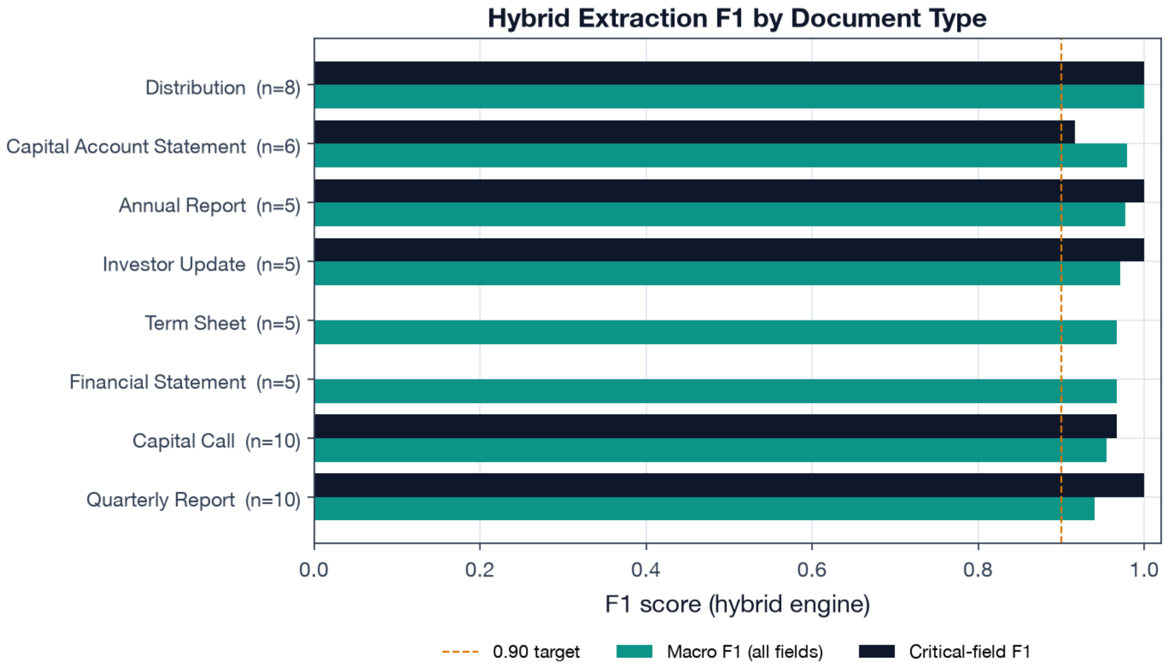
0.968 Critical-field F1

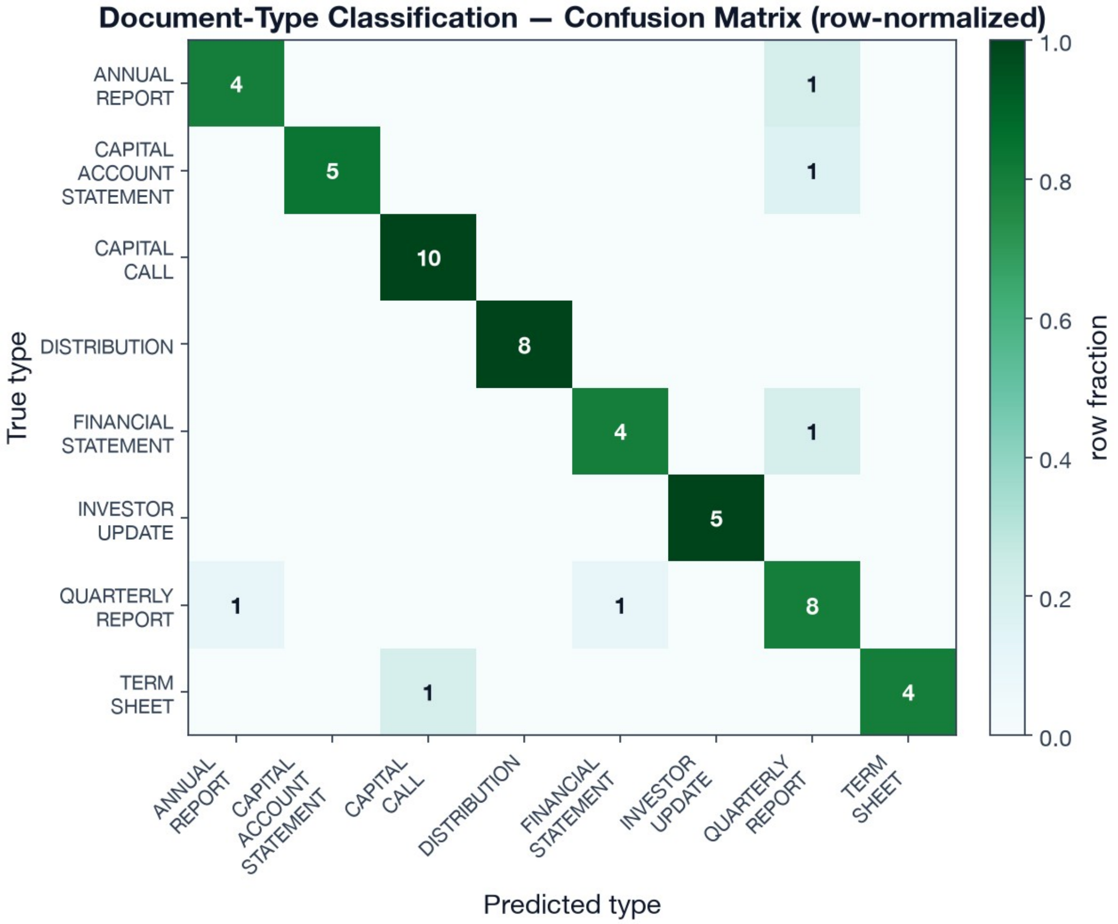
0.889 Classification acc

Document AI → tables/forms · LLM → narrative · Hybrid → merge precedence (merge.ts). Single engines fail in opposite directions.

DELIVERABLE 1 · EXTRACTION ACCURACY

Accuracy by document type & classification

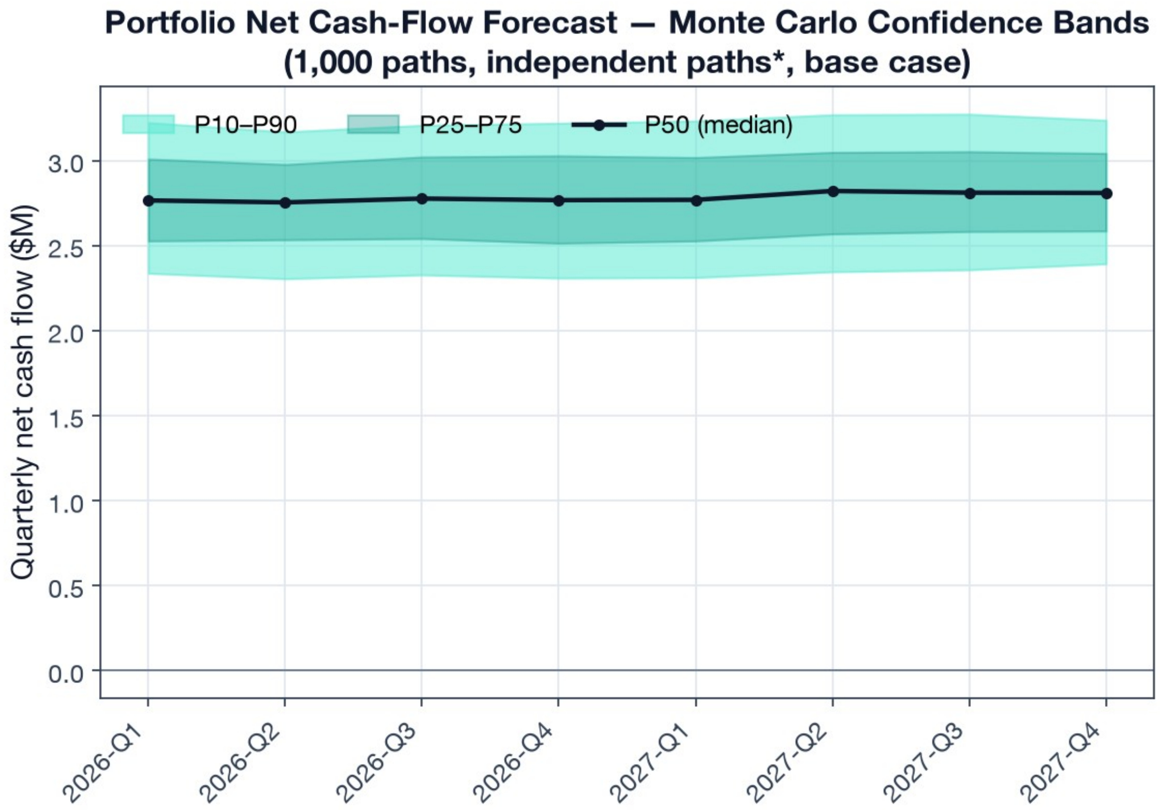


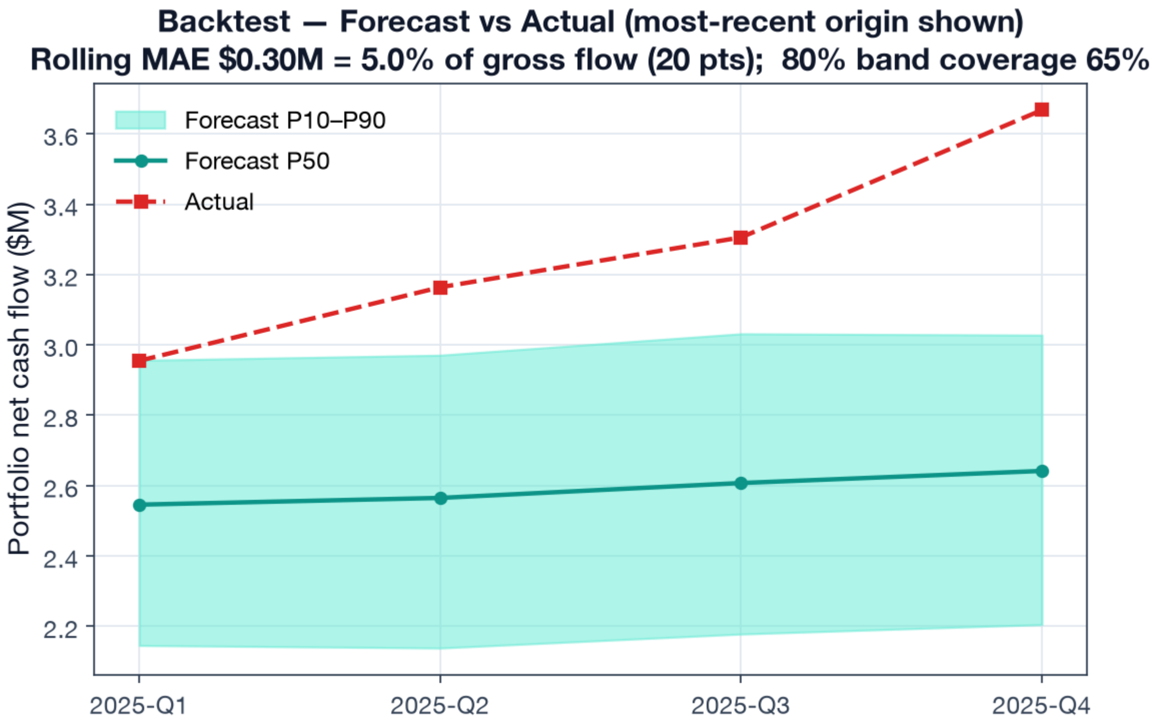


Hybrid F1 ≥ 0.94 for every type; classification confusions are only between adjacent periodic-report types.

DELIVERABLE 2 · FORECASTING

Monte Carlo forecast & backtest (MAE 5.0% < 15%)

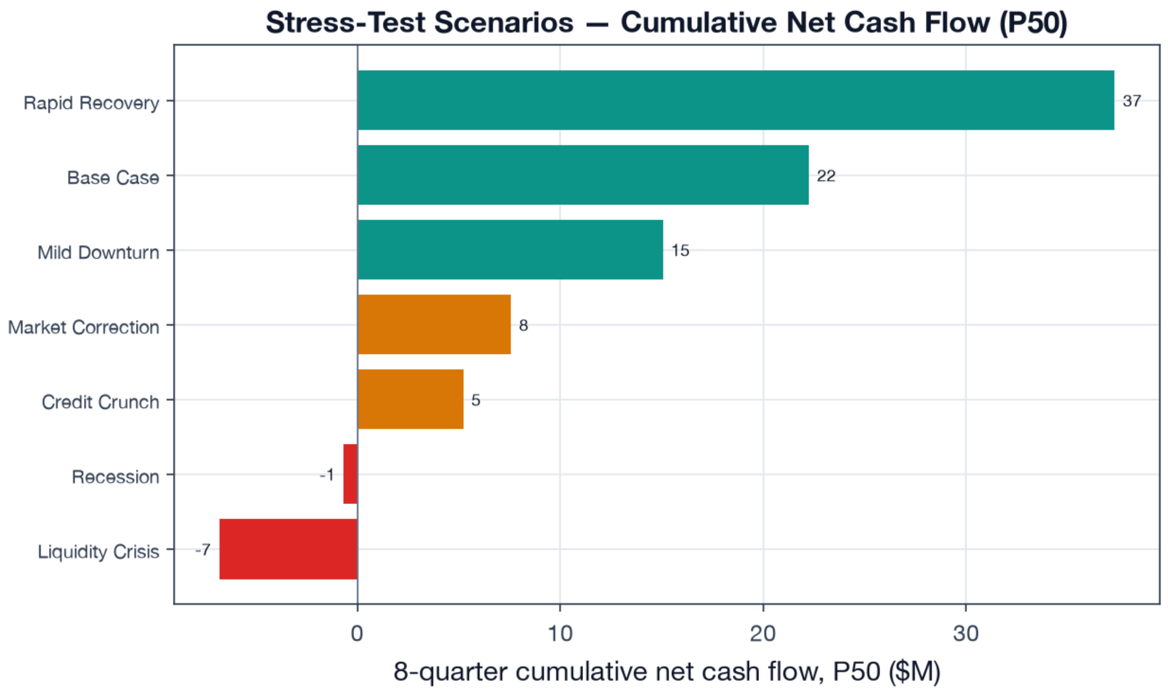


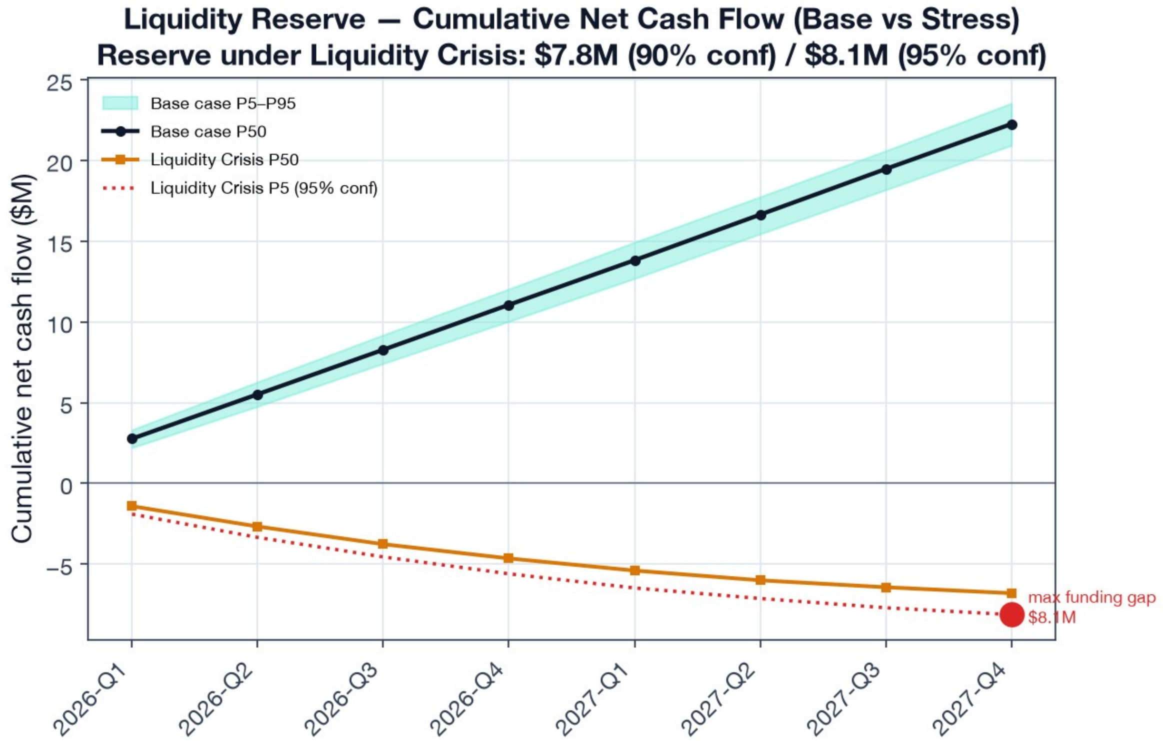


Engine is a line-for-line port of monte-carlo.ts. Rolling-origin backtest, 20 portfolio-quarter test points.

DELIVERABLE 2 · FORECASTING

Stress testing & liquidity reserve

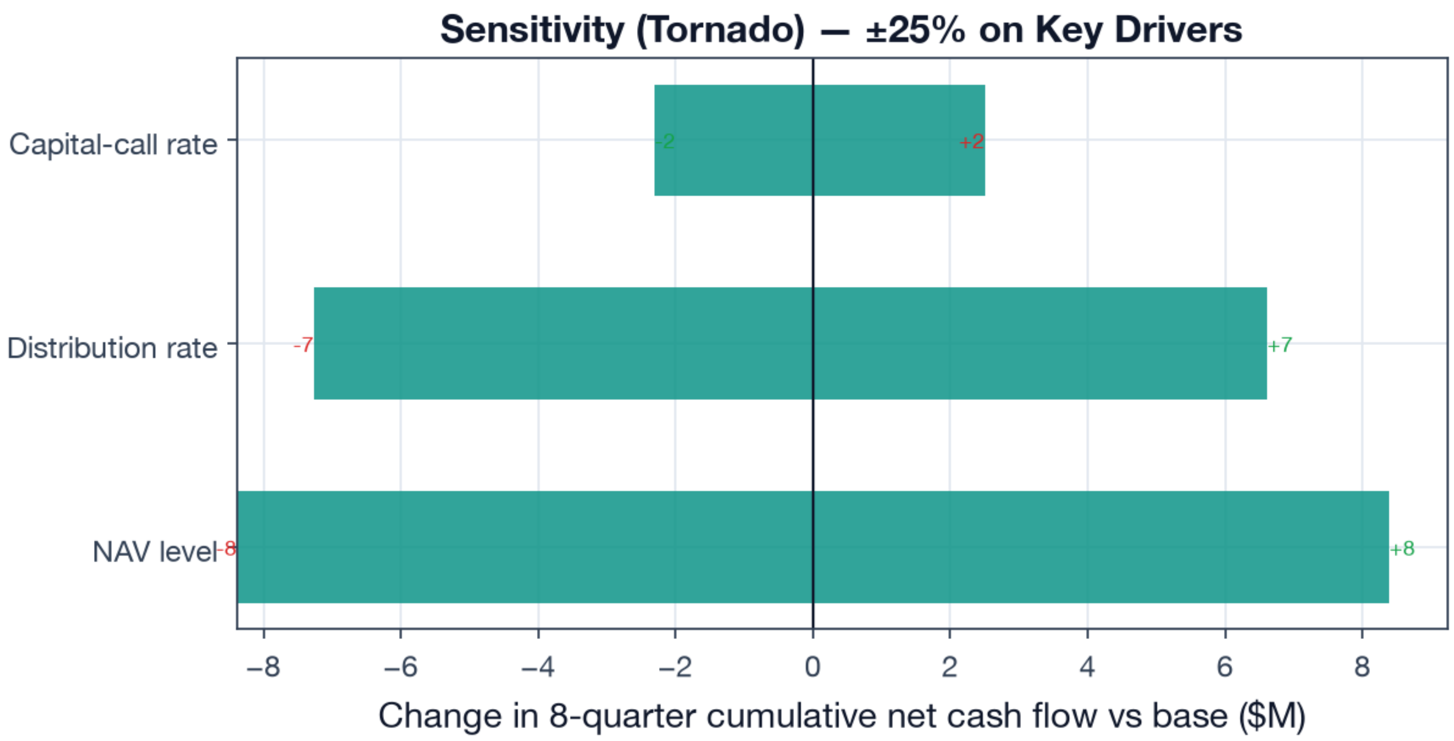




Self-funding in base case; ~$8.1M reserve (95% confidence) needed to survive a Liquidity Crisis.

DELIVERABLE 2 · FORECASTING

Sensitivity — what moves the forecast (±25%)



±$17M NAV level

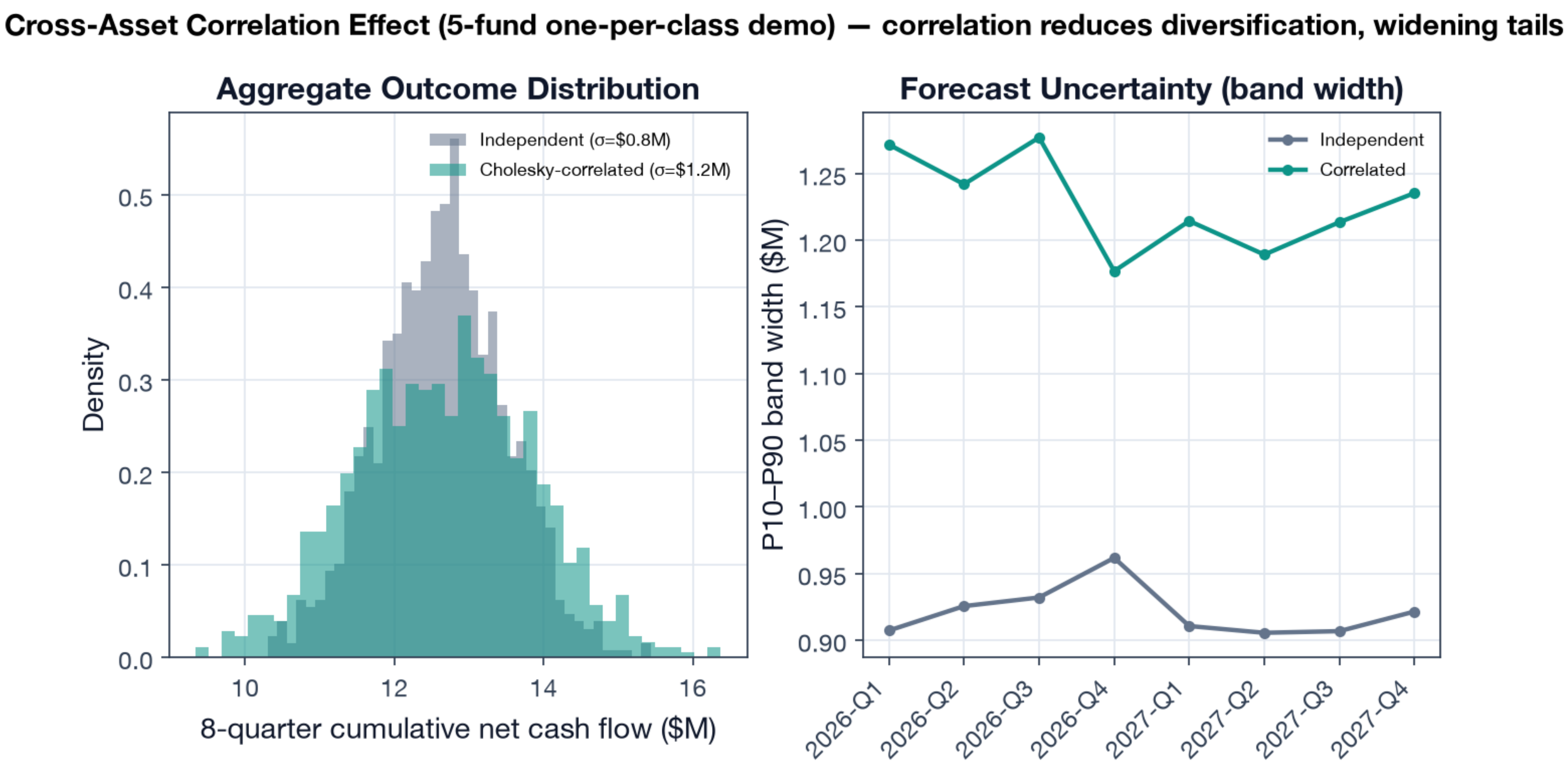
±$14M Distribution rate

±$5M Capital-call rate

NAV level and distribution pacing dominate; call timing matters ~3× less for a mostly-deployed book.

DELIVERABLE 2 · FORECASTING

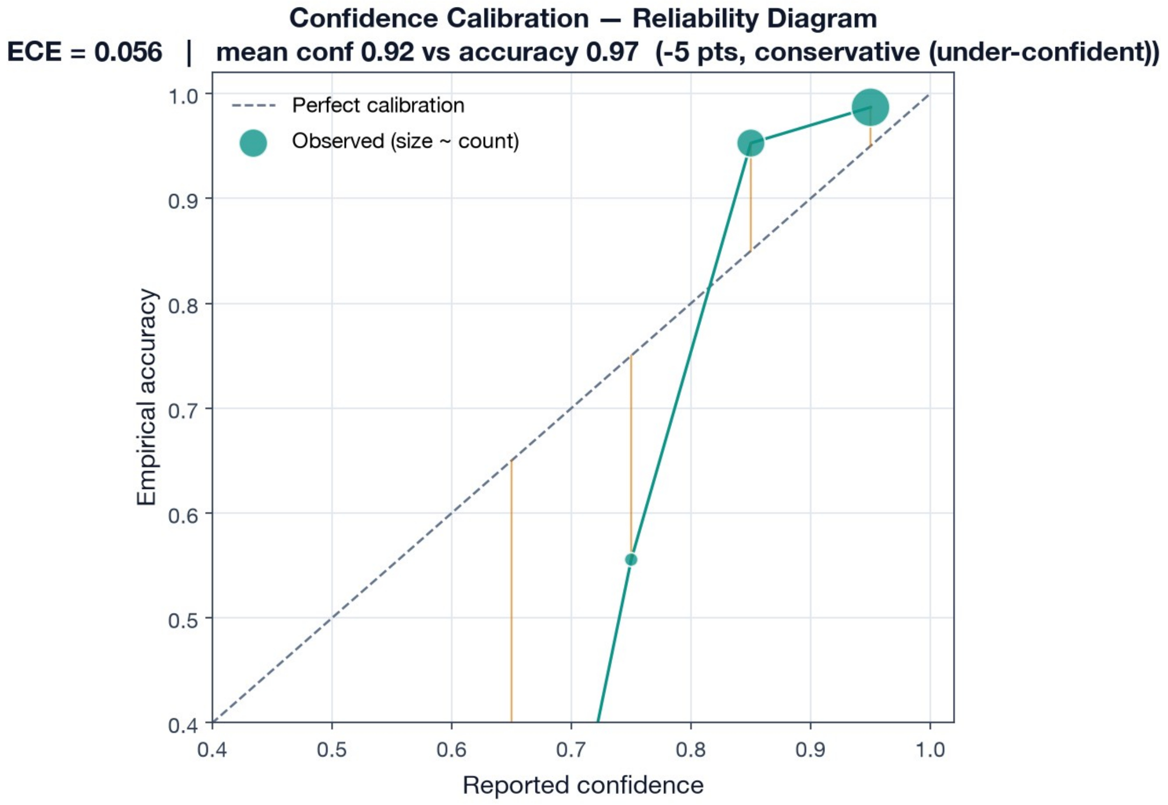
Cross-asset correlation widens the tails (+39% σ)

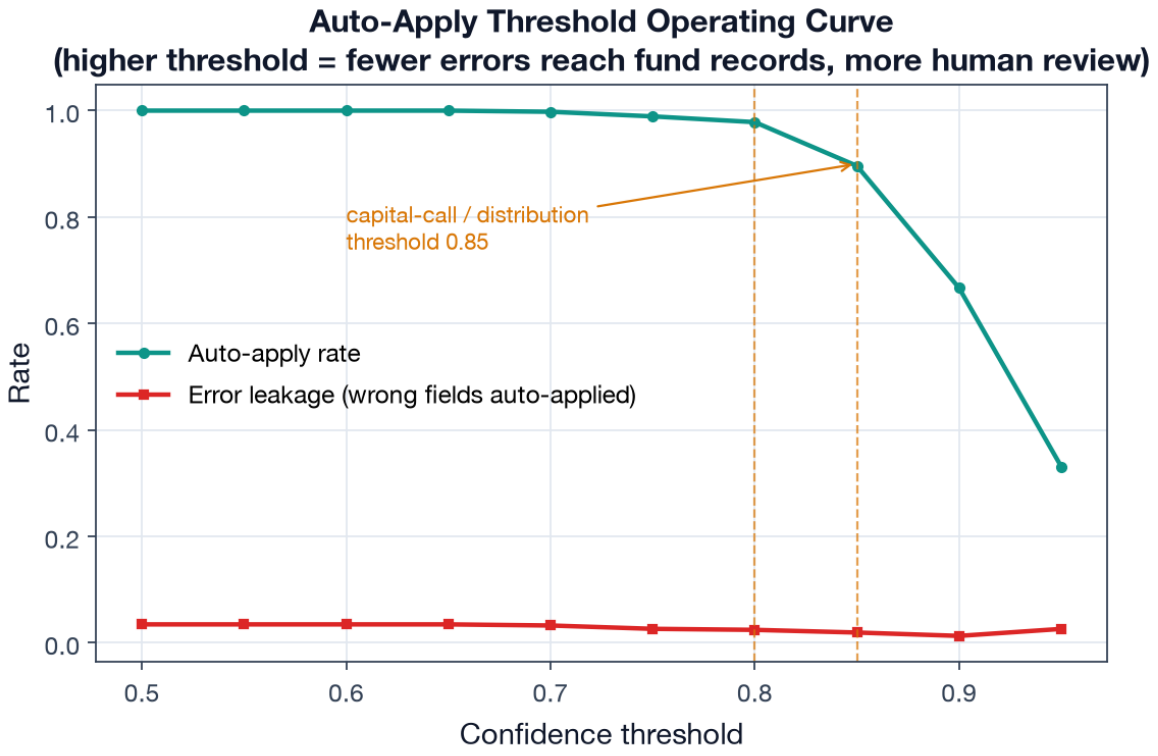


Cholesky correlation reduces diversification on a one-per-class demo — but no-ops on the full 16-fund book (singular matrix, monte-carlo.ts:336-341): a documented fix.

DELIVERABLE 3 · PIPELINE ANALYSIS

Confidence calibration & auto-apply thresholds

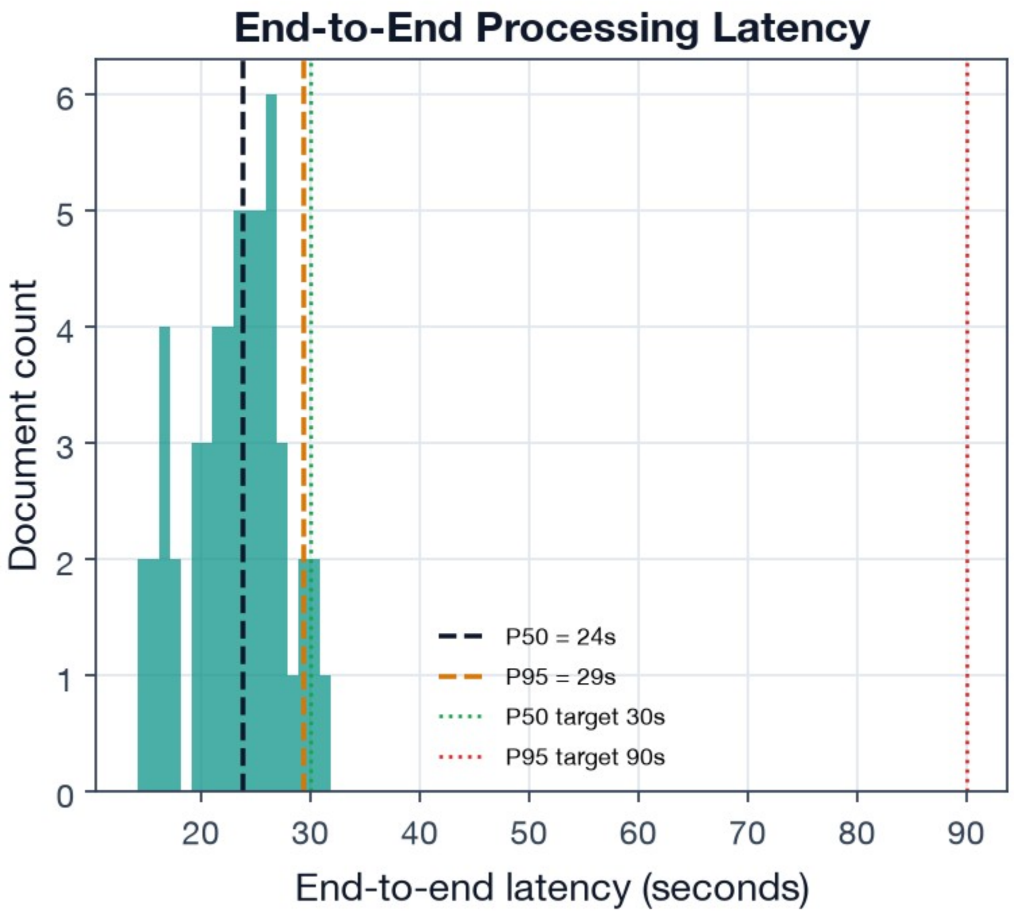


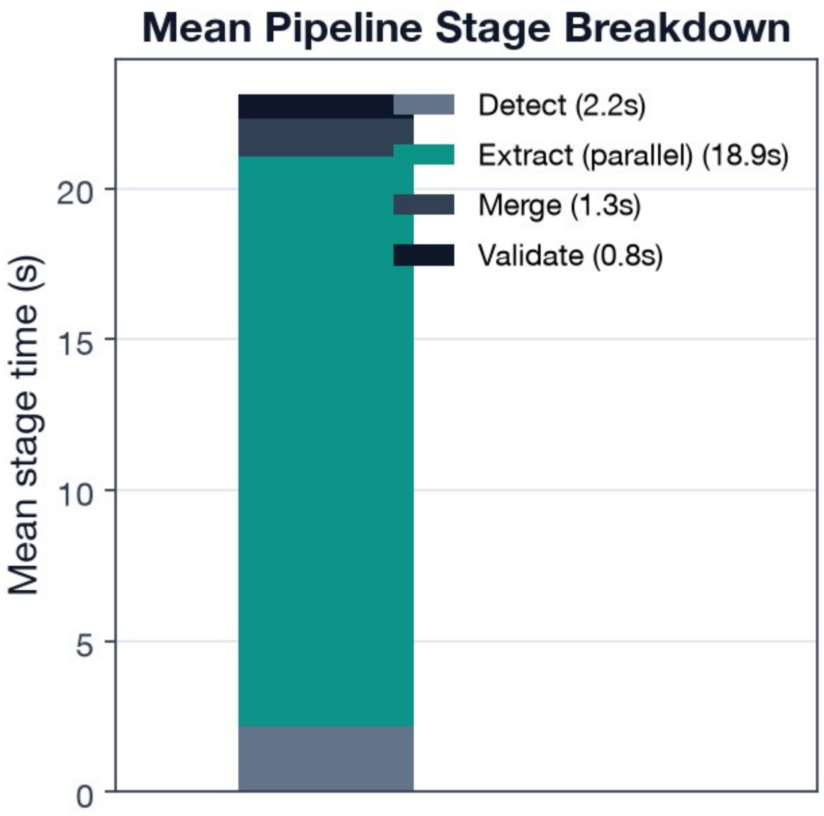


Conservatively calibrated (under-states reliability) → thresholds can be relaxed after a calibration pass.

DELIVERABLE 3 · PIPELINE ANALYSIS

Latency meets target: P50 24s / P95 29s (Metric 3)





Targets P50 < 30s / P95 < 90s — both met. Document AI + LLM run in parallel, so extraction cost is the slower engine, not the sum.

Conclusions

Architecture choices are evidence-backed.

Hybrid extraction and Monte Carlo each beat the simpler alternative on the metric that matters.

The system is honest about uncertainty.

Conservatively calibrated (ECE 0.056); classification errors are 'safe' adjacent-type confusions.

Two concrete fixes carry forward.

Widen forecast intervals (65% coverage of an 80% band); regularize the correlation matrix.

Production numbers are wired.

Every metric recomputes unchanged against live extractions / real cash flows — only inputs change.